

M Akshaya Varshinee

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in Akshaya Varshinee M

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📁 LeetCode

📁 Codechef

Education

Chennai Institute of Technology, Chennai

Bachelor of Technology in Artificial Intelligence and Data Science

September 2024 – May 2028

CGPA: 8.50

Experience

Cognizant Technology Solutions - Agentic AI Intern

April 21, 2026 – Present

- Building industry-oriented Agentic AI and RAG applications using Microsoft Azure AI services and cloud-native architectures.
- Working with Azure services such as Azure Foundry, Azure Containers, Azure Functions, and cloud deployment pipelines.
- Collaborating in a team environment to design scalable AI solutions for real-world business use cases.
- Implementing backend logic, API integrations, and data processing pipelines for AI-powered applications.
- Exploring optimization strategies for latency, token usage, and production cost efficiency in LLM-based systems.

Chennai Institute of Technology - COE Intern Drug Repurposing Project using Agentic AI

Nov 1, 2025 – Nov 28, 2025

- Worked on an individual project exploring drug repurposing using agentic AI. Designed two agent pipelines (drug analysis and disease analysis) that automatically retrieve biomedical data from multiple APIs including ChEMBL, OpenTargets, and BindingDB. Implemented using OpenAI Agents SDK and FastAPI to orchestrate tool-based information retrieval and evidence aggregation.
- Implemented tool-based agent workflows enabling automated biomedical knowledge retrieval and analysis.

Projects

Offline Multimodal RAG System (Ongoing)

LangChain, Qdrant, Multimodal RAG

- Developing an offline-first multimodal retrieval augmented generation system supporting text, document, and image-based queries.
- Implemented local embedding generation, vector indexing with Qdrant, and similarity-based document retrieval.
- Designed a retrieval pipeline enabling local inference without cloud dependency.

Befunge-93 Esoteric Coding Contest Platform React, Node.js, PostgreSQL, BullMQ, Redis, k6 - [GitHub Link](#) Dec 2025 – Jan 2026

- Designed and developed a high-concurrency competitive programming platform for the **Befunge-93** esoteric language, supporting **200+ simultaneous participants** during live contests.
- Implemented an **asynchronous judging architecture** using **BullMQ** and Redis-backed job queues to process code submissions without blocking the main Node.js event loop.
- Built a **worker-based execution pipeline** where background workers fetch submissions from the queue, execute them using a Befunge-93 interpreter, and return results to the grading service.
- Developed backend services for **submission handling, job scheduling, result aggregation, and leaderboard updates** to support concurrent contest participants.
- Integrated **PostgreSQL** for persistent storage of submissions, contest metadata, scoring results and leaderboard generation.
- Conducted load testing using k6 with 400 concurrent users, achieving 0% HTTP failure rate under peak contest traffic.

Skills

AI / LLM Systems: OpenAI Agents SDK, CrewAI, LangChain (Basic), MCP, A2A, Retrieval-Augmented Generation (RAG)

Vector Databases: ChromaDB, Qdrant (basic)

Cloud: Microsoft Azure (Azure OpenAI, Azure AI Search, Azure Functions)

Programming: Python, C++, JavaScript, C, Java (Basic)

Web & Backend: FastAPI, Node.js, Express.js, React.js, HTML, CSS

Data & Databases: NumPy, Pandas, Matplotlib, Seaborn, Oracle, PostgreSQL, MySQL, MongoDB

Tools: Git, GitHub, Postman, k6, Grafana

Relevant Coursework: DSA, DBMS, OOPs, Web Development, OS, Computer Networks

Achievements

TCS CodeVita Season 13 – Round 1 Qualifier JETRO Talent for Japan 2025 – Finalist IIT Madras Esolang Contest – Finalist

IEEE DebugX (Intra-Department Event) – Second Runner-Up. H@CIT Codeathon 2025 & 2026 – Finalist

Volunteering

Technical Volunteer – Stranger Codes (Befunge-93 Contest), TALOS 5.0 (AI&DS Department Technical Symposium)

Contributed to the organisation of an esolang coding contest by developing the competition website, Befunge-93 syntax documentation, and designing problem solutions. Assisted with backend system design focused on asynchronous submission handling, queues, rate limiting, and database connection pooling. Provided on-site technical support during the event by clarifying language-specific queries and guidance for identifying bugs in participant's esolang code.